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Student Enrollment & Course Management System (SECMS)

INTRODUCTION:

- The **Student Enrollment & Course Management System (SECMS)** is a software application designed to manage student enrollment and course administration in educational institutions such as colleges, universities, and training centers.
- This system helps automate the process of student registration, course enrollment, and academic record management. It replaces manual paperwork with a digital system, making data management faster, more accurate, and more efficient.

Project Flow:

Milestone 1: Salesforce Developer Account Creation:

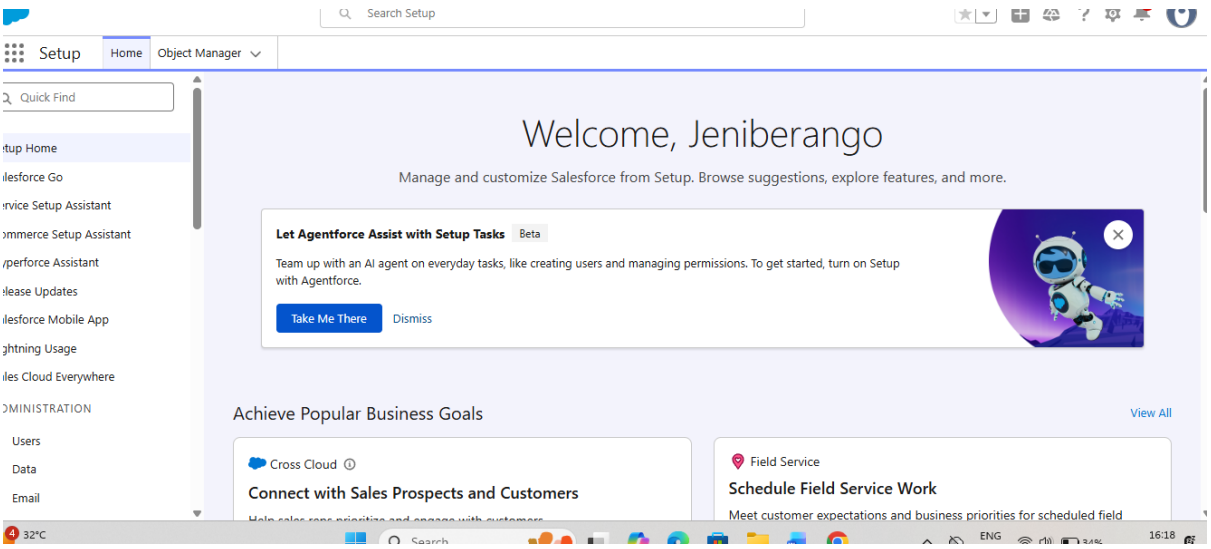
- Salesforce is a cloud-based Customer Relationship Management (CRM) platform

that allows developers to build, test, and deploy applications. To start developing applications on Salesforce, a **Salesforc Developer Account** is required. This account provides access to the Salesforce Developer Environment where users can create custom objects, workflows, and applications

steps to Create a Salesforce Developer Account

1. Open a web browser and go to the Salesforce Developer website.
2. Click on the **Sign Up** option to create a new account.
3. Fill in the required details such as:
    - First Name
    - Last Name
    - Email Address
    - Company Name
    - Role
    - Country
4. Choose a unique **Username** for the account (usually in email format).
5. Accept the terms and conditions.
6. Click the **Sign Me Up** button to submit the registration form.
7. A verification email will be sent to the registered email address.

8. Open the email and click the **verification link**.
9. Set a password and security question to activate the account.
10. After activation, log in to the Salesforce Developer Environment.



Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

- In Salesforce, **custom objects** are used to store and manage data specific to an application. For the **Student Enrollment & Course Management System (SECMS)**, custom objects are created to represent the core entities involved in the system.
- The main custom objects created for this project are **Student, Course, Enrollment, and Instructor**. These objects help organize and

manage information related to students, courses, instructors, and course enrollments.

Steps to Create Custom Objects in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Click on the **Setup** icon.
3. In the **Object Manager**, click **Create**.
4. Select **Custom Object**.
5. Enter the required details such as:
   - Label
   - Plural Label
   - Object Name
   - Record Name
6. Enable options like **Allow Reports**, **Track Field History**, and **Allow Search** if required.
7. Click **Save** to create the object.
8. Repeat the same process to create the remaining objects.

Custom Objects Created

1. Student Object

The **Student** object stores details about students enrolled in the institution.

Fields may include:

- Student Name
- Student ID
- Email

- Phone Number
- Date of Birth
- Address

Purpose:
To maintain a centralized database of all student information.

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2. Course Object

The **Course** object contains information about courses offered by the institution.
Fields may include:

- Course Name
- Course Code
- Course Duration
- Course Description
- Credits

Purpose:
To manage and organize all available courses.

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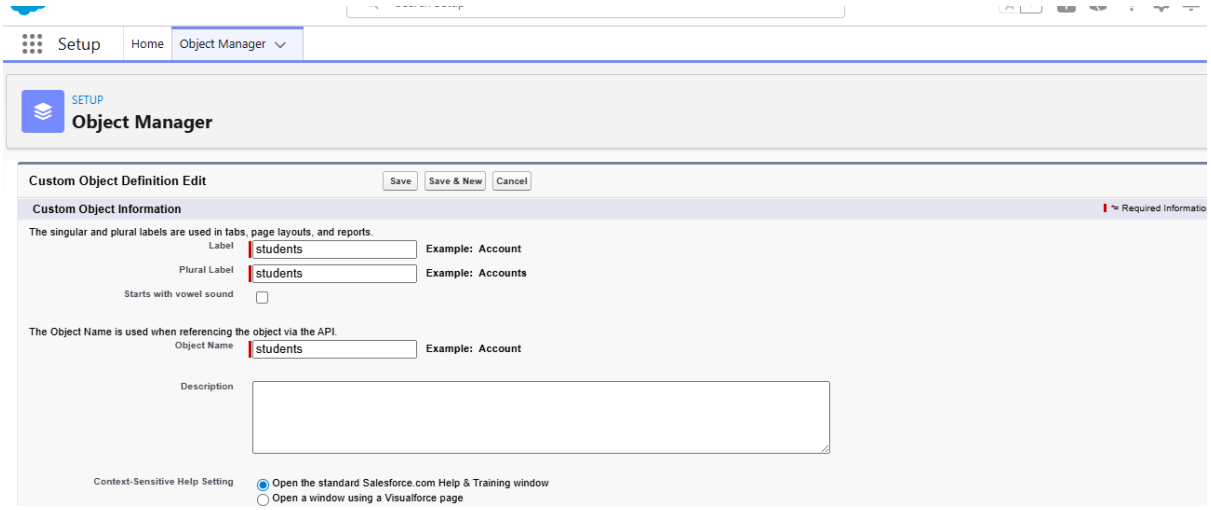
3. Instructor Object

The **Instructor** object stores details about faculty members who teach courses.
Fields may include:

- Instructor Name
- Instructor ID
- Email

- Phone Number
- Department



Milestone 3: Tabs Creation

- In Salesforce, **Tabs** are used to provide easy access to objects, records, and applications. Tabs allow users to quickly navigate between different parts of the system.
- For the **Student Enrollment & Course Management System (SECMS)**, tabs are created for the custom objects such as **Student, Course, Enrollment, and Instructor**. These tabs help users view, create, and manage records related to these objects efficiently.

Steps to Create Tabs in Salesforce

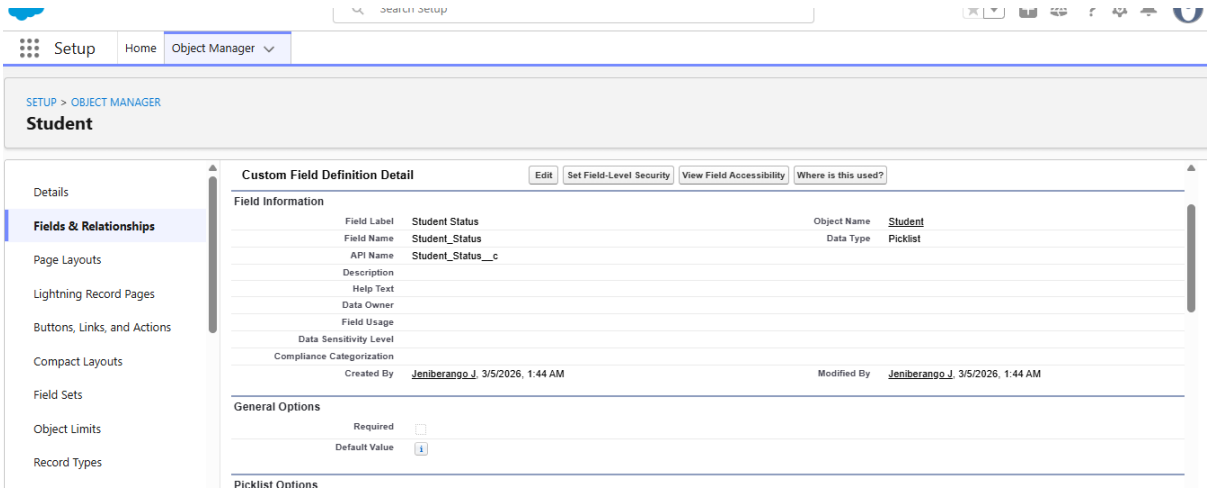
1. Log in to the **Salesforce Developer Account**.
2. Click on the **Setup** icon.
3. In the **Quick Find** search box, type **Tabs**.
4. Select **Tabs** under the **User Interface** section.

5. Click **New** under **Custom Object Tabs**.
6. Choose the object for which the tab needs to be created (e.g., Student).
7. Select a **Tab Style** (icon and color).
8. Click **Next**.
9. Assign the tab visibility to user profiles.
10. Add the tab to the required **Apps**.
11. Click **Save** to complete the process.
12. Repeat the same steps for the remaining objects.

Tabs Created

The following tabs were created for the system:

- **Student Tab** – To manage student records
- **Course Tab** – To manage course details
- **Instructor Tab** – To manage instructor information
- **Enrollment Tab** – To manage student course enrollments



Milestone 4: Lightning App Creation

- A **Lightning App** in Salesforce is used to group related objects, tabs, and features into a single application. It provides a user-friendly interface that allows users to access all necessary components of a system from one place.
- For the **Student Enrollment & Course Management System (SECMS)**, a Lightning App is created to combine the custom objects such as **Student, Course, Instructor, and Enrollment** into one application. This makes it easier for users to manage student enrollment and course information efficiently

Steps to Create a Lightning App in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Click on the **Setup** icon.
3. In the **Quick Find** box, type **App Manager**.
4. Select **App Manager** from the results.
5. Click the **New Lightning App** button.
6. Enter the **App Name** (e.g., Student Enrollment System).
7. Choose an **App Logo** and description if required.
8. Click **Next**.
9. Select **Navigation Style** (Standard Navigation or Console Navigation).
10. Add the required **Navigation Items** such as:

- Student

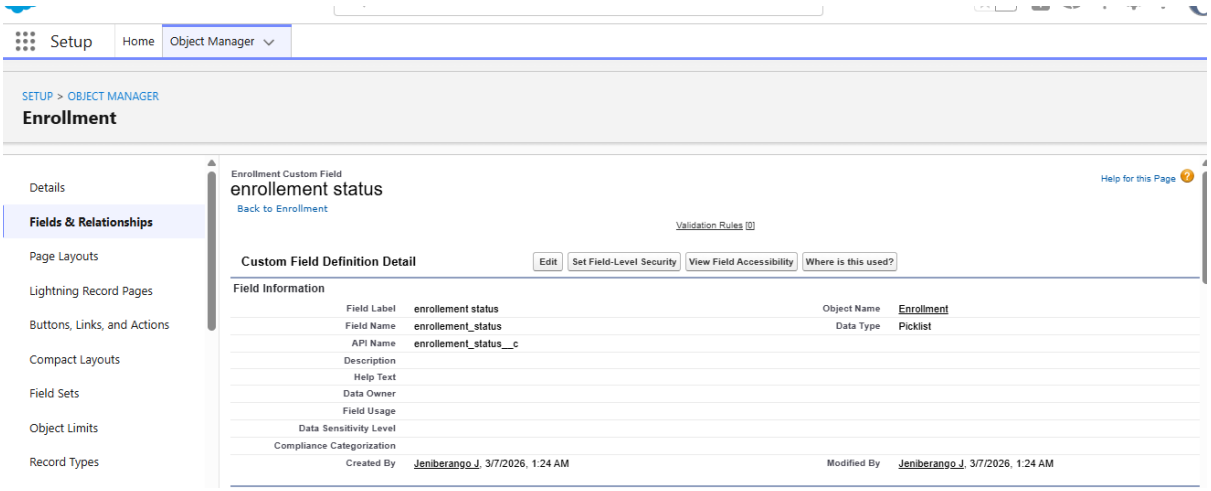
- Course
- Instructor
- Enrollment

11. Assign **User Profiles** that can access the app.
12. Review the settings and click **Save & Finish**.

Components Included in the Lightning App

The Lightning App for SECMS includes the following components:

- **Student Tab** – Manage student records
- **Course Tab** – Manage course details
- **Instructor Tab** – Manage instructor information
- **Enrollment Tab** – Manage student enrollments



Milestone 5 – Validation Rules:

- **Validation Rules** in Salesforce are used to ensure that the data entered into the system meets certain conditions or requirements. These rules help maintain data accuracy and prevent

users from saving incorrect or incomplete information.

- In the **Student Enrollment & Course Management System (SECMS)**, validation rules are created to control the data entered for students, courses, instructors, and enrollments.

Steps to Create Validation Rules in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup**.
3. Click on **Object Manager**.
4. Select the object where the validation rule needs to be created (e.g., Student).
5. Click **Validation Rules**.
6. Click **New** to create a new rule.
7. Enter the **Rule Name** and **Description**.
8. Define the **Rule Formula** based on the required condition.
9. Enter an **Error Message** that will be displayed if the rule is violated.
10. Choose where the error message should appear (Field or Top of Page).
11. Click **Save** to activate the rule.

Validation Rules Implemented

1. Student Email Validation

Ensures that the student email field is not left blank.

Example Rule:

- If the Email field is empty, the record cannot be saved.

Error Message:

"Student Email is required."

---

2. Course Duration Validation

Ensures that the course duration is greater than zero.

Example Rule:

- If the course duration is less than or equal to zero, the record cannot be saved.

Error Message:

"Course duration must be greater than zero."

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3. Instructor Phone Number Validation

Ensures that the phone number contains a valid number of digits.

Example Rule:

- Phone number must contain a valid format.

Error Message:

"Enter a valid phone number."

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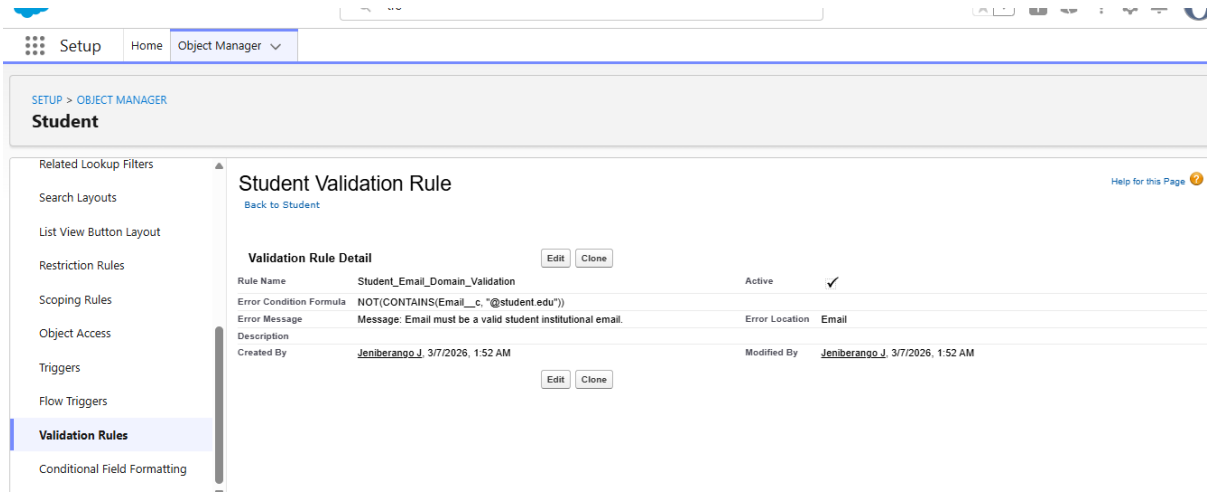
4. Enrollment Date Validation

Ensures that the enrollment date is not set in the past.

Error Message:

"Enrollment date must be today or a future date."



Milestone 6: Create Relationships

- In Salesforce, **relationships** are used to connect different objects so that related data can be managed efficiently. Relationships help link records between objects and allow users to view associated information easily.
- For the **Student Enrollment & Course Management System (SECMS)**, relationships are created between the **Student, Course, Instructor, and Enrollment** objects. These relationships help track which student is enrolled in which course and which instructor is responsible for teaching the course.

Types of Relationships Used

The following relationship types are commonly used in Salesforce:

1. **Lookup Relationship**
   - Creates a loose connection between two objects.

- Records can exist independently.

2. **Master-Detail Relationship**
   - Creates a strong relationship between objects.
   - The detail record depends on the master record.

   For this project, **Lookup Relationships** are mainly used.

Steps to Create Relationships in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup**.
3. Click **Object Manager**.
4. Select the object where the relationship will be created (e.g., Enrollment).
5. Click **Fields & Relationships**.
6. Click **New**.
7. Choose the relationship type (Lookup Relationship).
8. Select the related object (e.g., Student or Course).
9. Enter the field label and details.
10. Set field-level security and page layout settings.
11. Click **Save**.

Relationships Created in SECMS

1. Student – Enrollment Relationship

- One **Student** can have multiple **Enrollments**.

- A Lookup relationship is created from **Enrollment** to **Student**.

  Purpose:

  To track which courses a student has enrolled in.

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2. Course – Enrollment Relationship

- One **Course** can have multiple **Enrollments**.
- A Lookup relationship is created from **Enrollment** to **Course**.

Purpose:

To track students enrolled in a specific course.

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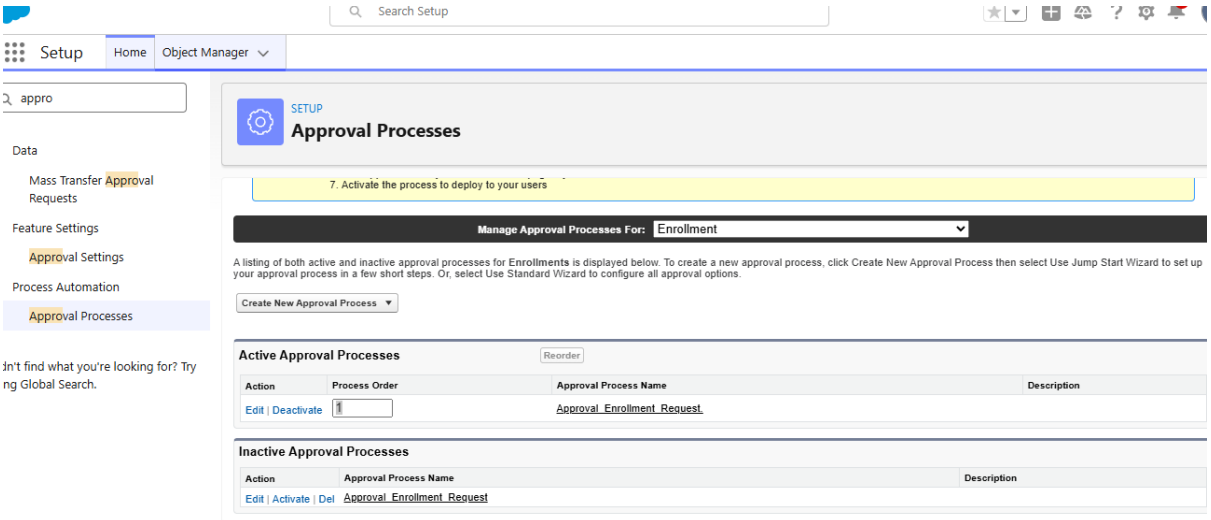
3. Instructor – Course Relationship

- One **Instructor** can teach multiple **Courses**.
- A Lookup relationship is created between **Course** and **Instructor**.

Purpose:

To assign instructors to courses.



Milestone 7: Page Layout Customization

- **Page Layout Customization** in Salesforce allows administrators to control how fields, related lists, and sections appear on a record page. It helps organize information in a clear and user-friendly manner so that users can easily view and manage records.
- In the **Student Enrollment & Course Management System (SECMS)**, page layouts are customized for objects such as **Student, Course, Instructor, and Enrollment**. This ensures that important fields are displayed properly and improves the usability of the system.

Steps to Customize Page Layout in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup**.
3. Click on **Object Manager**.
4. Select the object you want to customize (e.g., Student).
5. Click **Page Layouts**.
6. Select the existing layout or click **New** to create a new one.
7. Drag and drop fields, sections, and related lists into the layout.
8. Organize fields into logical sections such as Personal Details or Course Information.

9. Save the layout after making the changes.

Page Layout Customization for Objects

1. Student Object Layout

Sections created:

- Student Information
- Contact Details
- Enrollment Details

Fields included:

- Student Name
- Student ID
- Email
- Phone Number
- Date of Birth
- Address

---

2. Course Object Layout

Sections created:

- Course Details
- Instructor Information

Fields included:

- Course Name
- Course Code
- Course Duration
- Course Description
- Credits

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3. Instructor Object Layout

Sections created:

- Instructor Information
- Contact Details

Fields included:

- Instructor Name
- Instructor ID
- Email
- Phone Number
- Department

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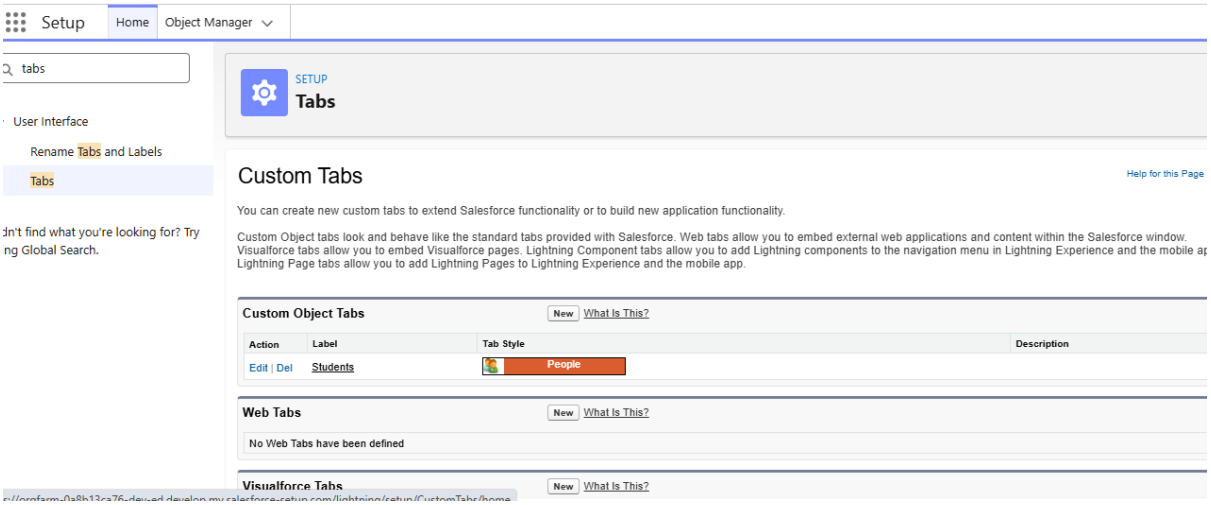
4. Enrollment Object Layout

Sections created:

- Enrollment Information
- Student and Course Details

Fields included:

- Student Name
- Course Name
- Enrollment Date
- Enrollment Status



Milestone 8: Validation Rules

- **Validation Rules** in Salesforce are used to ensure that the data entered into the system follows specific conditions before a record is saved. These rules help maintain data accuracy and prevent users from entering incorrect or incomplete information.
- In the **Student Enrollment & Course Management System (SECMS)**, validation rules are applied to objects such as **Student, Course, Instructor, and Enrollment** to ensure that the data entered is valid and consistent.

Steps to Create Validation Rules in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup**.
3. Click on **Object Manager**.
4. Select the object where the validation rule needs to be applied (e.g., Student).

5. Click **Validation Rules**.
6. Click **New** to create a new rule.
7. Enter the **Rule Name** and **Description**.
8. Write the **Rule Formula** based on the validation condition.
9. Enter the **Error Message** that will appear when the rule condition is not satisfied.
10. Choose where the error message should appear (field level or top of the page).
11. Click **Save** to activate the validation rule.

Validation Rules Implemented

1. Student Email Required
Ensures that the student email field cannot be left blank.
Error Message:
"Student email is required."

---

2. Course Duration Validation
Ensures that the course duration must be greater than zero.
Error Message:
"Course duration must be greater than zero."

---

3. Instructor Phone Number Format

Ensures that the instructor phone number follows a valid format and contains the correct number of digits.
Error Message:
“Please enter a valid phone number.”

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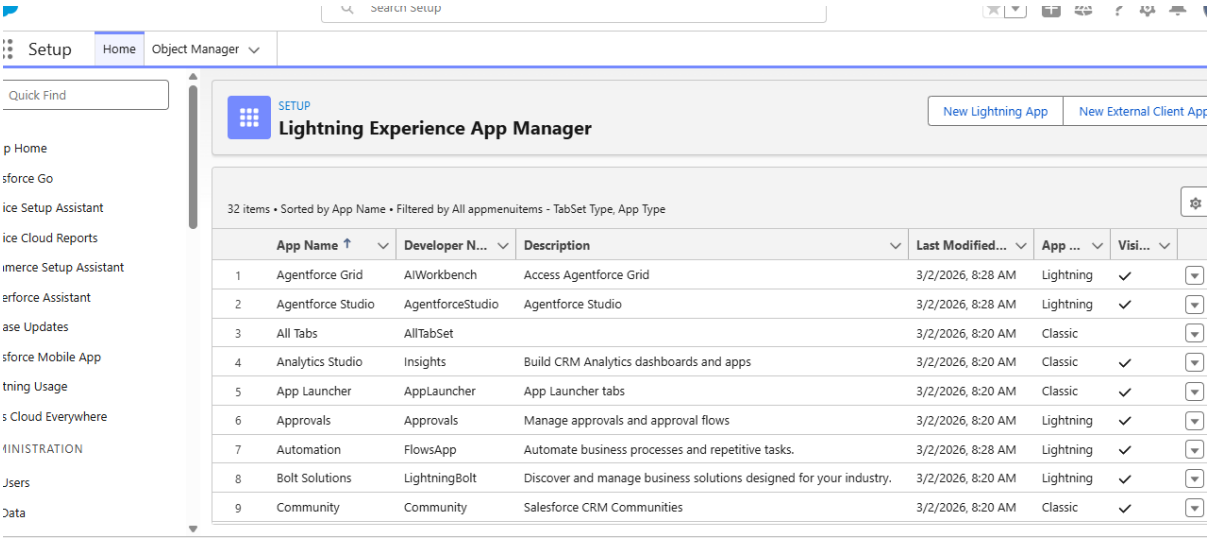
4. Enrollment Date Validation

Ensures that the enrollment date is not earlier than the current date.
Error Message:
“Enrollment date cannot be in the past.”



- In Salesforce, **Page Layouts** control the organization and visibility of fields, sections, related lists, and custom buttons on an object’s record page. Once the layouts are customized, users can create and manage **records** efficiently.
  - For the **Student Enrollment & Course Management System (SECMS)**, page layouts are finalized for the **Student, Course, Instructor,**

and Enrollment objects, and sample records are created to test the system functionality.

Steps for Page Layout Customization and Record Creation

Step 1: Customize Page Layouts

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup** → **Object Manager**.
3. Select the object to customize (e.g., Student).
4. Click **Page Layouts** and select the layout to edit.
5. Organize fields, related lists, and sections logically.
6. Save the changes.

Step 2: Create Records

1. Navigate to the **App** created (e.g., Student Enrollment System).
2. Click on the **Tab** for the object (e.g., Student).
3. Click **New** to create a record.
4. Fill in the required fields.
5. Click **Save** to add the record to the database.
6. Repeat the process for **Course, Instructor, and Enrollment** objects.

Sample Records Created

1. Student Records

Student Name	Student ID	Email	Phone	DOB
John Doe	STU001	john@example.com	9876543210	2002-05-12
Jane Smith	STU002	jane@example.com	9876543211	2003-08-20

2. Course Records

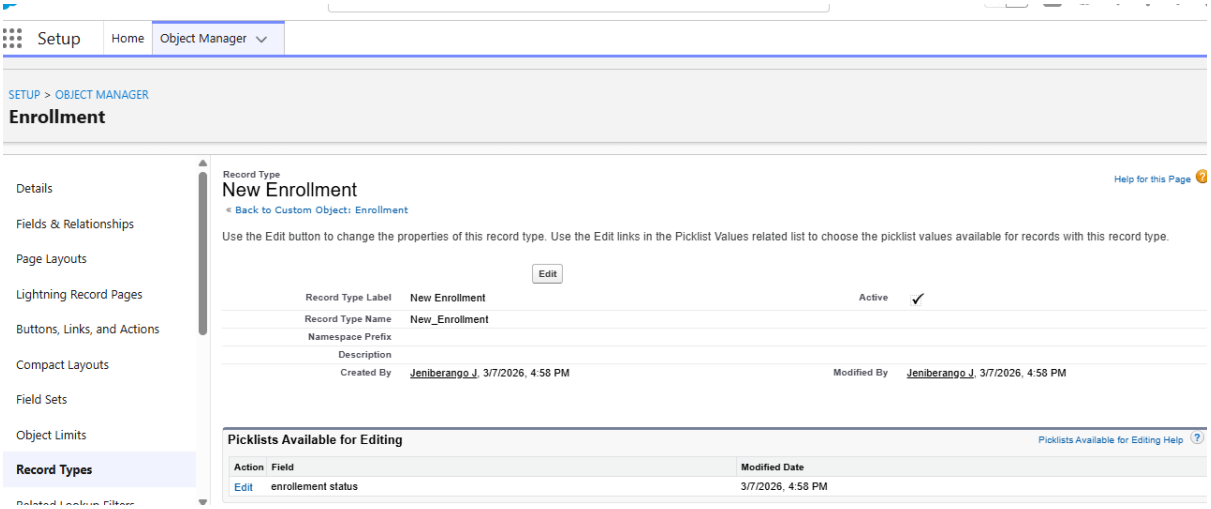
Course Name	Course Code	Duration	Credits
Computer Science	CS101	6 Months	4
Mathematics	MATH201	4 Months	3

3. Instructor Records

Instructor Name	Instructor ID	Email	Department
Alice Brown	INS001	alice@university.com	CS
Bob White	INS002	bob@university.com	Math

4. Enrollment Records

Student Name	Course Name	Enrollment Date	Status
John Doe	Computer Science	2026-03-01	Active
Jane Smith	Mathematics	2026-03-02	Active



Milestone 10: Approval Process

- The **Approval Process** in Salesforce automates the approval of records, ensuring that certain data is reviewed and approved by designated users before it becomes final. It is commonly used to enforce business rules and maintain data quality.
- In the **Student Enrollment & Course Management System (SECMS)**, the approval process can be applied to **Enrollment records**, ensuring that student course enrollments are reviewed and approved by an administrator or faculty before they are confirmed.

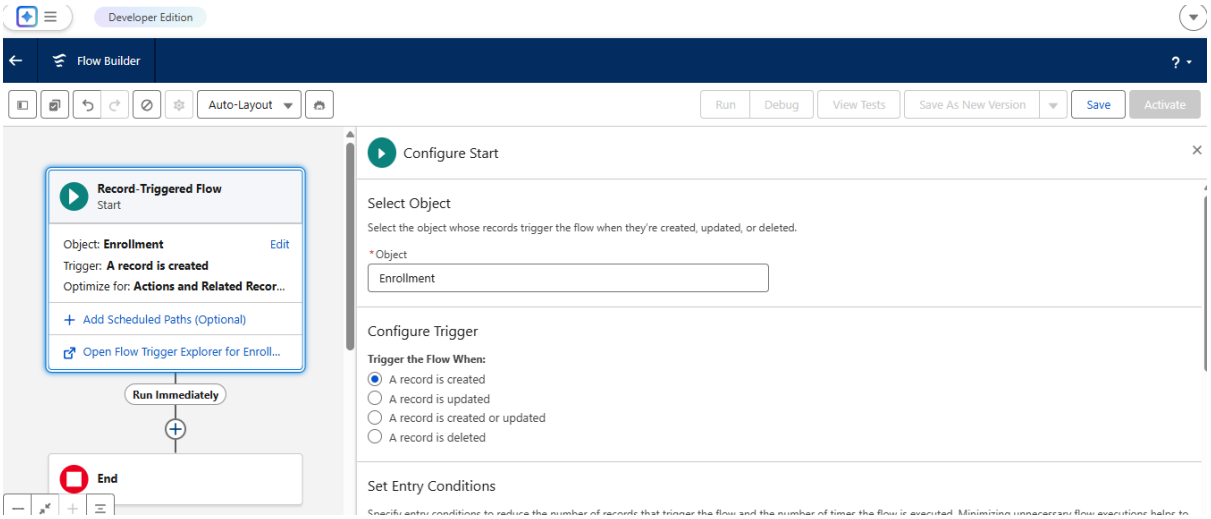
Steps to Create an Approval Process in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Go to **Setup** → **Object Manager**.
3. Select the object to create an approval process for (e.g., **Enrollment**).
4. Click **Approval Processes**.
5. Click **Create New Approval Process** and select **Use Standard Setup Wizard**.
6. Enter the **Name** and **Unique Name** for the approval process (e.g., "Enrollment Approval").
7. Define the **Entry Criteria**:
   - For example, only records with Enrollment Status = "Pending" enter the approval process.
8. Select **Approver**:
   - Assign to a specific user, role, or manager (e.g., Admin or Faculty).
9. Set **Initial Submission Actions**:
   - For example, lock the record to prevent edits during approval.
10. Set **Approval Actions**:
    - Change status to "Approved" and notify the student.
11. Set **Rejection Actions**:
    - Change status to "Rejected" and notify the student with a reason.

12.Activate the approval process.

Example Approval Process: Enrollment

Step	Action
Submission	Student submits enrollment request for a course.
Entry Criteria	Enrollment Status = “Pending”
Approver	Admin or Faculty assigned to approve enrollments.
Approval Action	Change status to “Approved”, notify student via email.
Rejection Action	Change status to “Rejected”, notify student via email with reason.



Milestone 11: Reports

- Reports in Salesforce are used to organize, summarize, and analyze data stored in objects. They provide insights into operations, helping administrators, faculty, and students make informed decisions.
- In the Student Enrollment & Course Management System (SECMS), reports are created to track student enrollments, course details, instructor assignments, and other academic data.

Steps to Create Reports in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Click on the **App Launcher** and select **Reports**.
3. Click **New Report**.
4. Choose the **Report Type** (e.g., Students, Courses, Enrollments, Instructors).
5. Click **Continue**.
6. Use the **Filters** to select the data to be displayed:
    - Example: Show only active enrollments or courses in the current semester.
7. Drag and drop fields to the report layout to display required information.
8. Use **Grouping** to organize data by student, course, or instructor.
9. Add **Summaries** for totals, counts, or averages.
10. Click **Save & Run** to generate the report.

11. Name the report and select a folder to save it in.

Sample Reports for SECMS

1. Student Enrollment Report

- Displays all students and the courses they are enrolled in.
- Fields: Student Name, Course Name, Enrollment Date, Status
- Grouped by: Course Name

2. Course Enrollment Summary

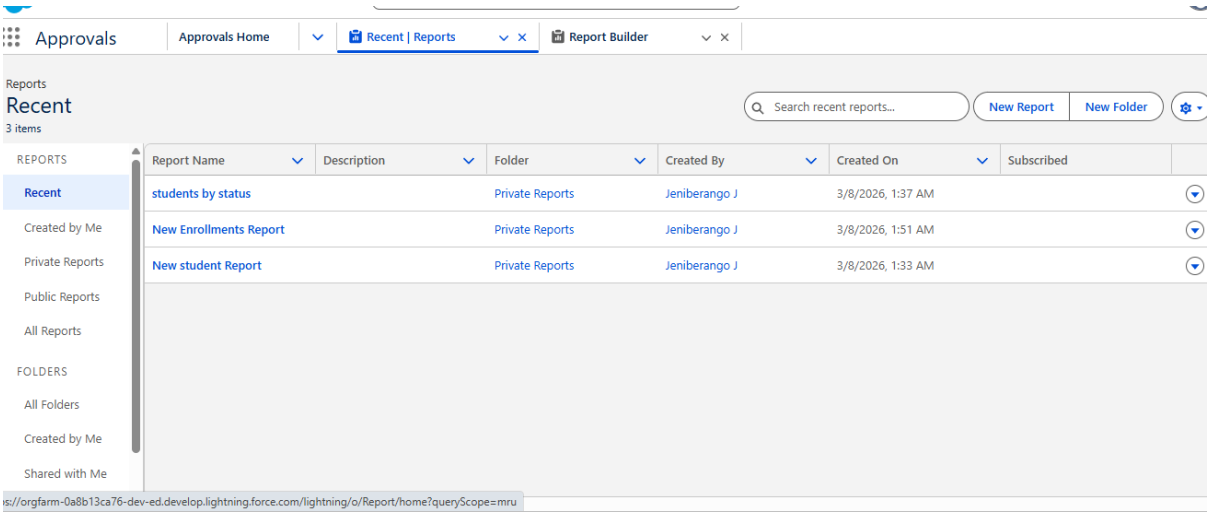
- Provides the total number of students enrolled per course.
- Fields: Course Name, Total Enrollments
- Summarizes: Count of students per course

3. Instructor Assignment Report

- Shows which courses are assigned to which instructors.
- Fields: Instructor Name, Course Name, Course Duration

4. Active Enrollments Report

- Displays all enrollments currently active.
- Fields: Student Name, Course Name, Enrollment Date, Status



Milestone12: Dashboards

- **Dashboards** in Salesforce provide a visual representation of data from multiple reports. They allow administrators, faculty, and management to quickly understand trends, track performance, and make data-driven decisions.
- In the **Student Enrollment & Course Management System (SECMS)**, dashboards are created to summarize student enrollments, course participation, and instructor assignments for easy monitoring and reporting.

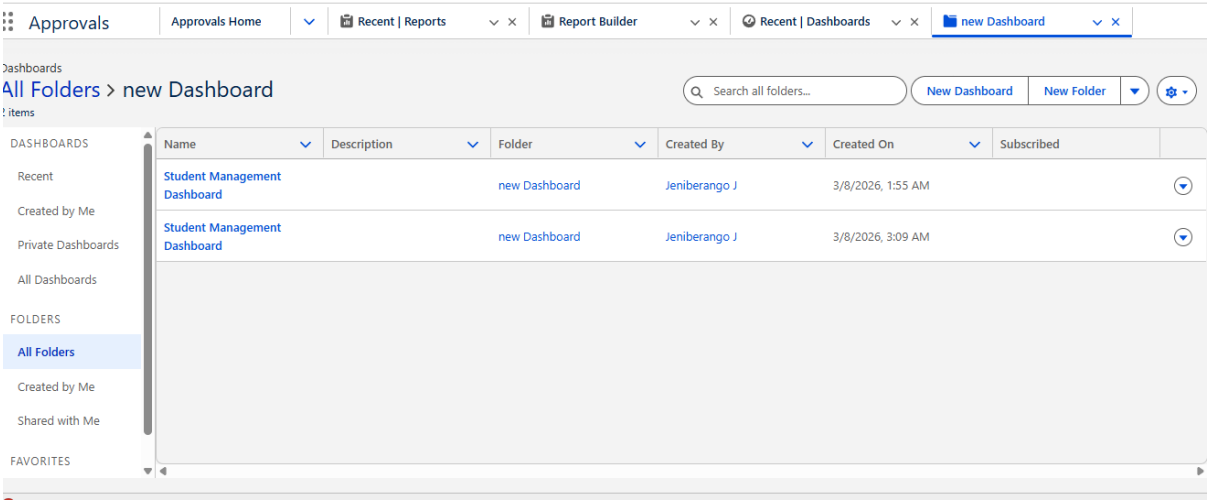
Steps to Create Dashboards in Salesforce

1. Log in to the **Salesforce Developer Account**.
2. Click on the **App Launcher** and select **Dashboards**.
3. Click **New Dashboard**.
4. Enter the **Dashboard Name** (e.g., “Student Enrollment Dashboard”) and select a folder to save it.
5. Click **Create**.

6. Click **+Component** to add a chart, table, metric, or gauge.
7. Select the **source report** for the component (e.g., Student Enrollment Report).
8. Choose the **visualization type**: Bar Chart, Pie Chart, Donut, Table, Gauge, etc.
9. Configure the **display settings**, grouping, and filters.
10. Add additional components to show multiple reports in one dashboard.
11. Click **Save** and then **Done**.

Sample Dashboard Components for SECMS

1. **Total Enrollments by Course** – Bar chart showing number of students per course.
2. **Active vs. Pending Enrollments** – Donut chart showing enrollment status distribution.
3. **Instructor Course Assignments** – Table listing instructors and their assigned courses.
4. **Enrollment Trends** – Line chart showing enrollment changes over months.
5. **Top Courses by Student Count** – Gauge highlighting the courses with the highest enrollment.



Milestone 13: Security Setup

- **Security Setup** in Salesforce ensures that data is protected and only accessible to authorized users. It allows administrators to control who can view, create, edit, or delete records. Security setup is critical for protecting sensitive student, course, and enrollment information in the **Student Enrollment & Course Management System (SECMS)**.

1. Profiles

- Profiles define what a user can do within Salesforce.
- Permissions include object-level access, field-level access, and administrative privileges.
- Example for SECMS:
  - **Admin Profile:** Full access to all objects, reports, dashboards, and approval processes.
  - **Faculty Profile:** Access to courses, students, and enrollments assigned to them.

- **Student Profile:** Limited access to their own student records and enrolled courses.

2. Roles

- Roles define the hierarchy of users and record visibility.
- Higher roles can view and manage records owned by users in subordinate roles.
- Example:
  - Admin → Faculty → Student

3. Permission Sets

- Permission sets are used to grant additional privileges beyond a user's profile.
- Example: Faculty may get extra permissions to edit course materials without changing their profile.

4. Organization-Wide Defaults (OWD)

- OWD defines the baseline access level for all objects in the organization.
- Example for SECMS:
  - **Student Object:** Private – only the owner and authorized users can view records.
  - **Course Object:** Public Read-Only – all users can view courses but only admins can edit.
  - **Enrollment Object:** Controlled by Parent – access is based on the Student and Course records.

5. Sharing Rules

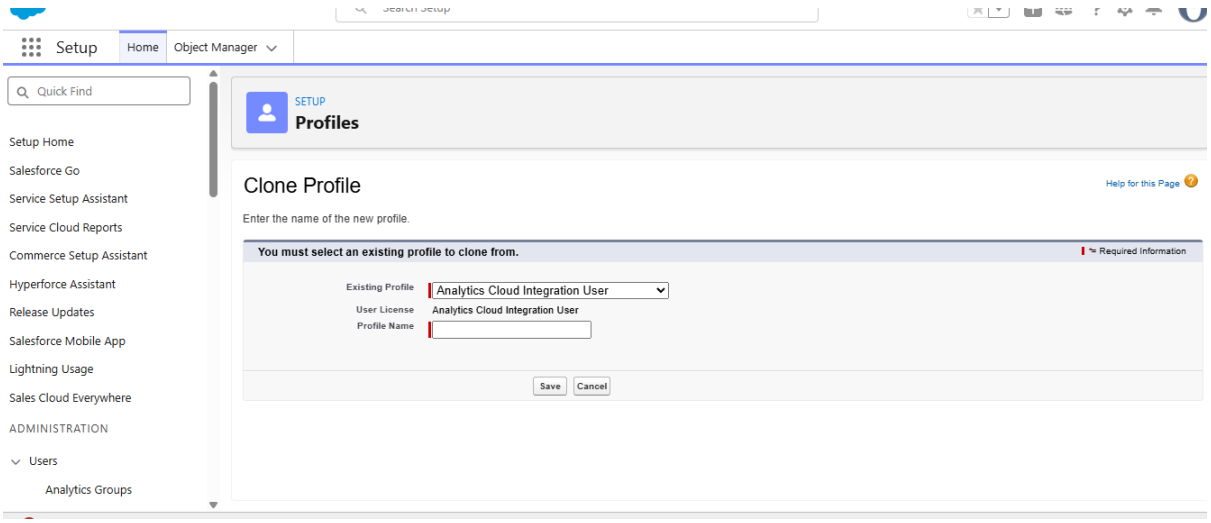
- Sharing rules grant additional record access based on criteria or ownership.
- Example:
    - Faculty can access all enrollments for courses assigned to them.

6. Field-Level Security

- Controls which fields are visible or editable for each profile.
- Example: Students cannot edit their own enrollment status field, but admins can.

Steps for Security Setup in SECMS

1. Go to **Setup → Users → Profiles** and configure object permissions for Admin, Faculty, and Student.
2. Configure **Roles** to define hierarchy and access levels.
3. Assign **Permission Sets** for additional privileges where needed.
4. Set **Organization-Wide Defaults (OWD)** for custom objects.
5. Create **Sharing Rules** to grant necessary access to records.
6. Set **Field-Level Security** for sensitive fields like Enrollment Status or Student Contact Details.
7. Test security setup by logging in as different users to verify access restrictions.



Milestone 14: Conclusion

The **Student Enrollment & Course Management System (SECMS)** is a comprehensive Salesforce-based solution designed to manage student registrations, course management, instructor assignments, and enrollments in an educational institution.

Through the project milestones, the system has been developed step by step, implementing objects, relationships, validation rules, page layouts, reports, dashboards, approval processes, and security features

1. **Salesforce Developer Setup:** Created a Salesforce Developer Account to develop the SECMS application.
2. **Custom Objects:** Developed key objects – Student, Course, Instructor, and Enrollment – to store academic data.

3. **Relationships:** Established lookup relationships between objects for proper data linkage.
4. **Tabs & Lightning App:** Created tabs and a Lightning App for easy access and navigation.
5. **Page Layouts & Records:** Customized layouts for better usability and created sample records for testing.
6. **Validation Rules:** Implemented validation rules to ensure data accuracy and integrity.
7. **Approval Process:** Automated enrollment approval workflow for better accountability.
8. **Reports & Dashboards:** Developed reports and dashboards to provide insights into students, courses, and enrollments.
9. **Security Setup:** Configured profiles, roles, permission sets, sharing rules, and field-level security to protect sensitive information.

Conclusion

- The **Student Enrollment & Course Management System (SECMS)** is a complete, secure, and automated solution that streamlines the enrollment and course management process in educational institutions.
- By leveraging Salesforce's features such as custom objects, relationships, workflows, validation rules, reports, dashboards, and security, SECMS ensures a highly efficient, accurate, and user-friendly system.